

MTH 310
Spring, 2026

Final Exam

Name: _____
Calculus III (Fall 2025)

Instructions. (200 points) Show work for credit. Calculator OK (but not useful).

- (18^{pts}) 1. Consider the line that passes through the point $(2, 0, 8)$ and is orthogonal to the plane $2x + 4y - 3z = 9$. Find the point at which this line intersects the plane.

- (18pts) **2.** Show that the limit $\lim_{(x,y) \rightarrow (0,0)} \frac{x^2 y}{x^4 + y^2}$ is zero along every line of the form $y = mx$ with $m \neq 0$. Just because the limit is zero along all lines, does that prove the limit is zero? Prove it is zero or demonstrate that it isn't.

- (18^{pts}) **3.** Suppose a mountain can be represented by the function $f(x, y) = x^2 + 2y^2 + xy - 2y$.
- (a) (9 pts) If you are on the mountain at the point $(0, 1, 0)$, in what direction should you move in order to decrease in elevation most quickly?
- (b) (9 pts) If you are at the point $(0, 1, 0)$ and decide to walk toward the point $(4, 4, 56)$, then what is the rate of change of elevation as you start your walk?

- (18^{pts}) 4. Give an equation for the tangent plane to the surface $x^2z^3 + 3y^3z = x^3 \cos(\pi y) + 9$ at the point $(1, -1, 1)$.

- (20^{pts}) **5.** Find all absolute maxima and minima of $f(x, y) = 2x^2 + 3y^2 - 4x - 5$ over the region $x^2 + y^2 \leq 4$. Don't forget to locate and check the critical points along the boundary.

(18^{pts}) **6.** Evaluate $\iiint_E z \, dV$ where E is the solid bounded above by $x^2 + y^2 + z^2 = 1$ and below by $z = \sqrt{x^2 + y^2}$.

(18^{pts}) **7.** Evaluate $\int_C x \, ds$ where C is the arc of the parabola $y = x^2$ from $(0, 0)$ to $(1, 1)$.

- (18^{pts}) **8.** Compute $\int_C (2x + \sin y) dx + (x \cos y - 6y) dy$, where C consists of line segments from $(0, 0)$ to $(2, 1)$ and from $(2, 1)$ to $(0, 2)$.

(18^{pts}) **9.** Let C be the circle of radius 5 centered at the origin traversed counterclockwise. Compute

$$\int_C (e^{\sin x} - y^3) dx + x^3 dy.$$

Hint: computing this directly isn't the way to go...

(18^{pts}) **10.** Let S be the surface parameterized by $\mathbf{r}(u, v) = \langle u \cos v, u \sin v, v \rangle$ for $0 \leq u \leq 2$, $0 \leq v \leq \pi$. Compute $\iint_S \sqrt{x^2 + y^2} dS$.

- (18^{pts}) **11.** Compute $\iint_S \mathbf{F} \cdot d\mathbf{S}$ where $\mathbf{F}(x, y, z) = \langle x^3, y^3, y^3 \rangle$ and S is the surface of the cylindrical solid bounded by $x^2 + y^2 = 1$, $z = 0$, $z = 4$, with outward orientation.